Directional Pedestrian Flow Analysis Using Edge Computing: An Educational Framework for AI-Driven Spatial Analytics

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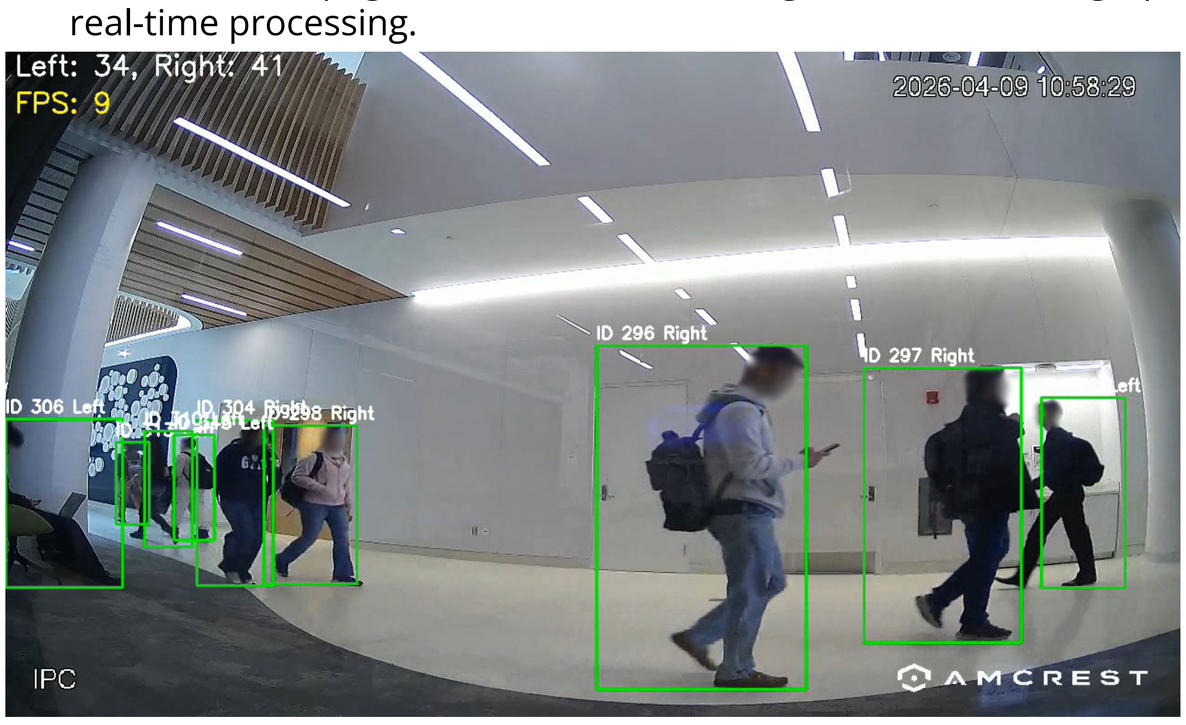
Abstract

Electronic Visualization Laboratory at the University of Illinois Chicago has been exploring edge computing applications through the SAGE ecosystem, particularly focusing on AI-driven spatial analytics. Traditional pedestrian tracking systems typically stream video to centralized servers for processing, raising privacy concerns and requiring substantial network bandwidth. We introduce a system that tracks walking directions in real time using NVIDIA Jetson Developer Kits. The AI process is local instead of in the cloud, then identifies movement but deletes all video to keep data private.

Data Processing

- Use YOLOv8s to find all people in an image and create a bounding box per person.
- Use BoT-SORT to track and keep the identity per person to re-identify if lost, creating an ID for each person.
- Once a person(ID) is no longer visible check first x-coordinate location and subtract it from last x-coordinate location.
- Record date, time, direction, and system telemetry to save and sent to the cloud via a REST-API.
- The cloud uses SQLite to store the data and use queries to send information to the front-end webpage.
- Front-end web pages show the visual storage of the data in a graph form for users to see real-time processing.



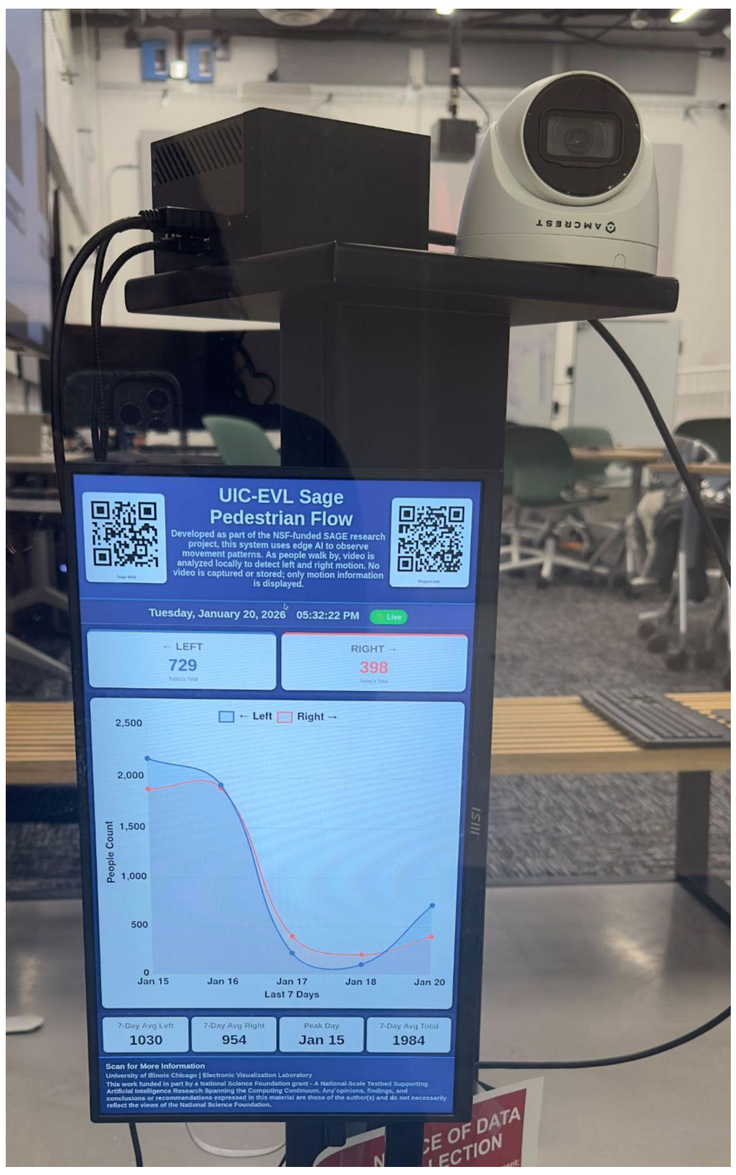

```
Date              : 2026-04-09
Time_Minute       : 14:28
avg_CPU_usage     : 20.58
avg_GPU_usage     : 1.6
avg_mem_usage     : 6674.2
avg_CPU_temp      : 49.72
avg_GPU_temp      : 45.11
direction_left    : 2
direction_right   : 3
```

System Usage

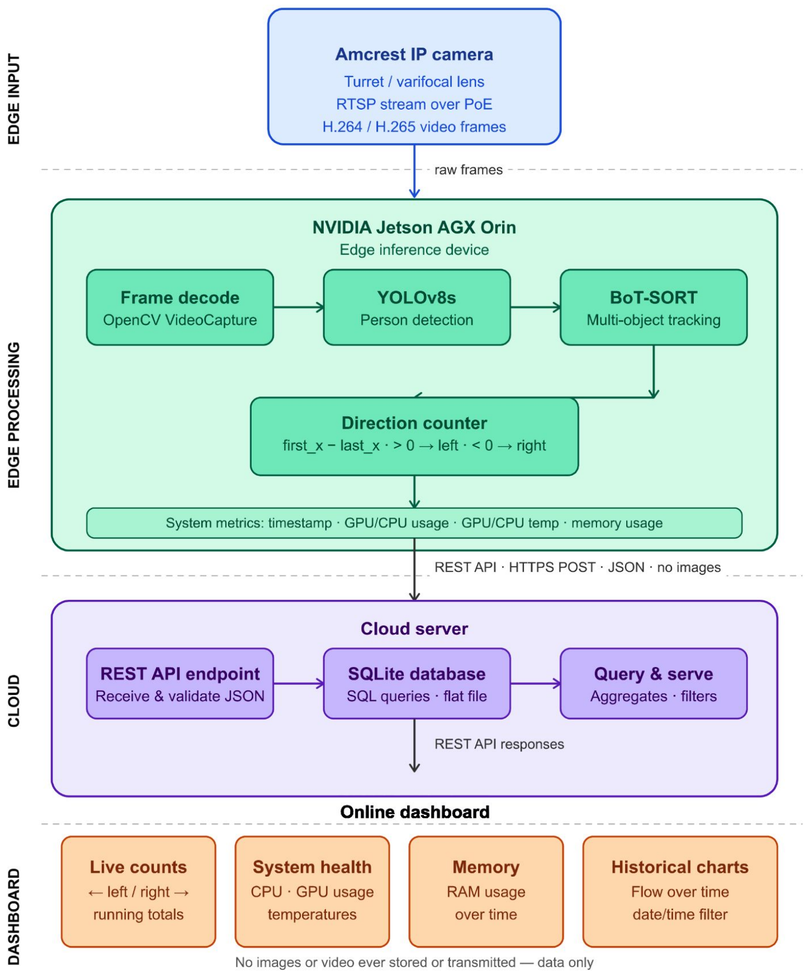
Avg CPU usage	: 25.55%
Avg GPU usage	: 32.07%
Avg Memory usage	: 7296.0 MB
Avg CPU temp	: 47.39°C
Avg GPU temp	: 42.55°C

Setup

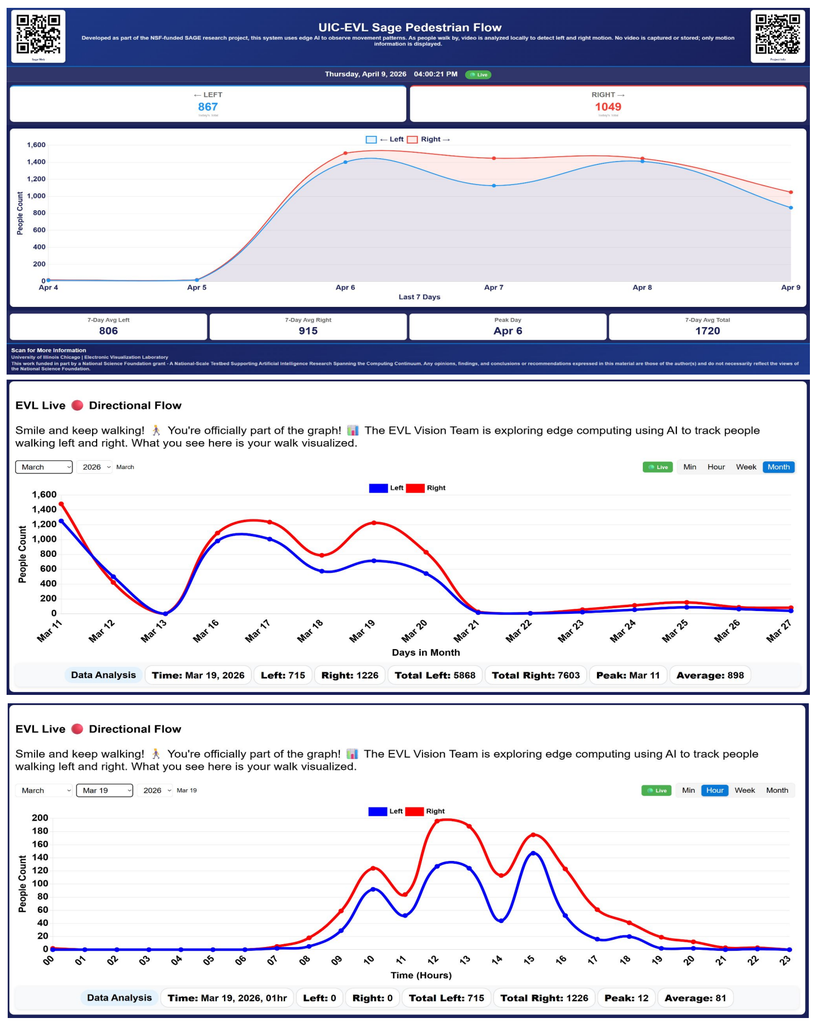
- NVIDIA Jetson AGX Orin
- GPU: NVaIDIA Ampere architecture
  - 2048 CUDA cores
  - 64 Tensor cores.
  - Max Frequency: 1.3 GHz
- CPU: 12-core Arm Cortex-A78AE v8.2 (64-bit)
  - Max Frequency: 2.2 GHz
- Memory: 64GB 256-bit LPDDR5



System Architecture



Results



Conclusion

- **End-to-End Pipeline:** Deployed a complete edge-to-cloud system for real-time directional pedestrian tracking.
- **Real-Time Performance:** Maintained sustained processing speeds of 15 FPS directly on edge hardware.
- **System Stability:** Hardware telemetry (CPU/GPU load, temperatures) remained safely within operational limits during prolonged inference.
- **Data Privacy:** Processed all video feeds locally; only anonymized numerical metadata is transmitted to the cloud.
- **Robust Tracking:** BoT-SORT successfully preserved pedestrian identities and features despite temporary camera occlusions.
- **Live Visualization:** REST API integration reliably populates a cloud dashboard for immediate remote monitoring of spatial flows.
- **Educational Impact:** Created an accessible framework to lower the barrier for students engaging with AI edge computing.

References & Acknowledgement

[1] C. Catlett *et al.*, "Hands-On Computer Science: The Array of Things Experimental Urban Instrument," *Computing in Science & Engineering*, vol. 24, no. 1, pp. 57-63, 2022.

[2] C. Catlett *et al.*, "Measuring Cities with Software-Defined Sensors," *IEEE Journal of Social Computing*, vol. 1, no. 1, pp. 14-27, 2020.

[3] P. Beckman *et al.*, "Harnessing the Computing Continuum for Programming Our World," in *Fog Computing: Theory and Practice*, Wiley, 2020.

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